Planning and Reasoning

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Agenda

- ReAct Framework
- Plan-and-Execute Framework
- Self Reflection

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Let’s begin the discussion by answering a few questions.

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Planning and Reasoning Quiz

What is the key cycle followed by ReAct agents ?

A. Plan → Execute → Measure → Report

B. Reason → Act → Observe → Repeat

C. Collect → Analyze → Conclude → Execute

D. Input → Process → Output → Feedback

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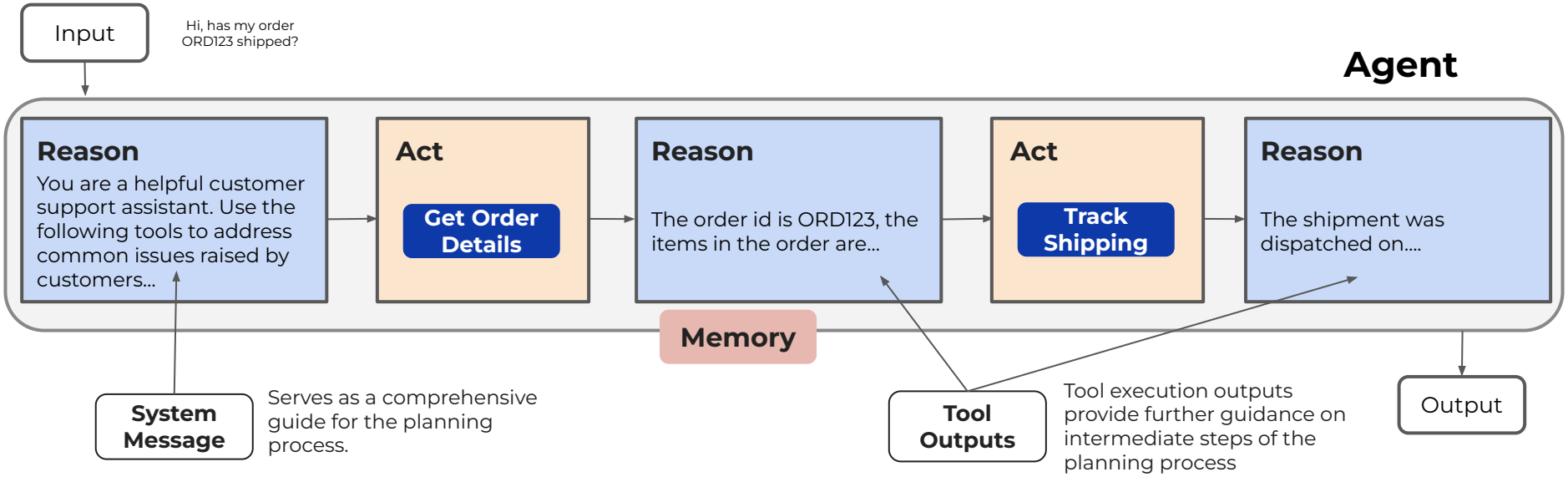
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ReAct Framework in AI Agents

In ReAct, the agent alternates between Reasoning and Acting.

Tool results feedback into the agent's memory, informing subsequent reasoning and actions until the final objective is reached and an output is generated.



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Planning and Reasoning Quiz

What is the defining feature of a Plan-and-Execute agent compared to ReAct?

A. It decides the next step only after each tool result and may revise direction continuously

B. It creates an explicit step-by-step plan upfront, then executes those steps

C. It is limited to a single tool call and must return an answer immediately after that call

D. It primarily depends on long-term memory to select actions rather than an explicit plan

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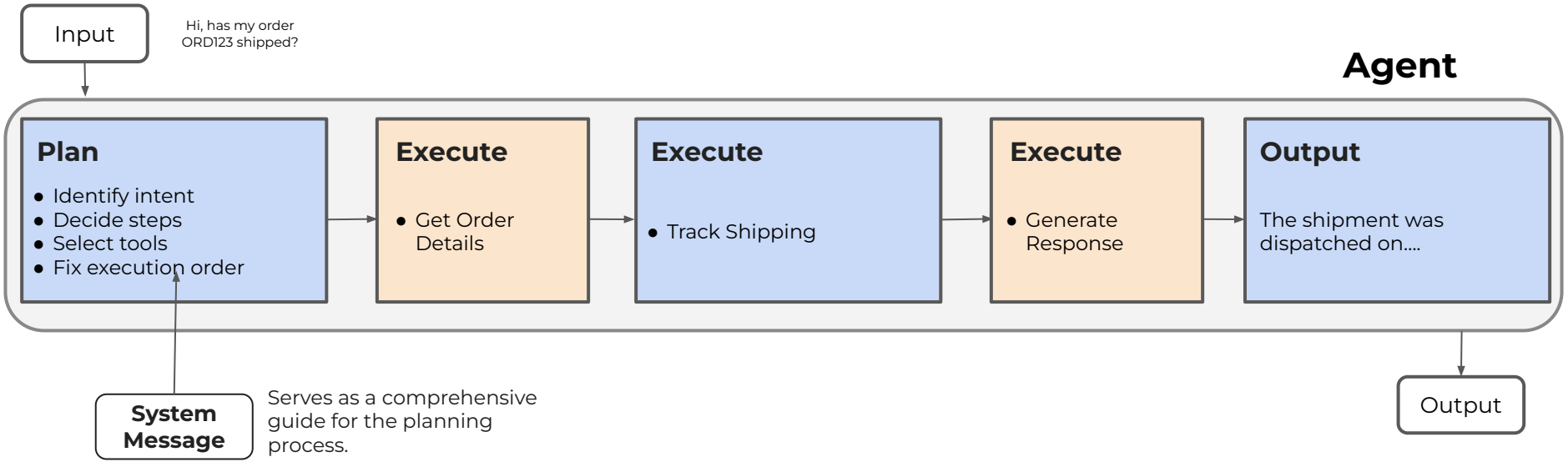
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Plan and Execute Framework in AI Agents

In the Plan-and-Execute approach, the agent first creates a complete execution plan before taking any action.

Once the plan is finalized, the agent executes each step sequentially using tools, without interleaving reasoning after every action.



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Planning and Reasoning Quiz

What is the primary purpose of self-reflection in AI agents?

A. To make the agent respond faster by skipping intermediate reasoning

B. To permanently store knowledge across sessions

C. To improve output quality by critiquing and revising the agent's own responses

D. To ensure the agent never uses external tools

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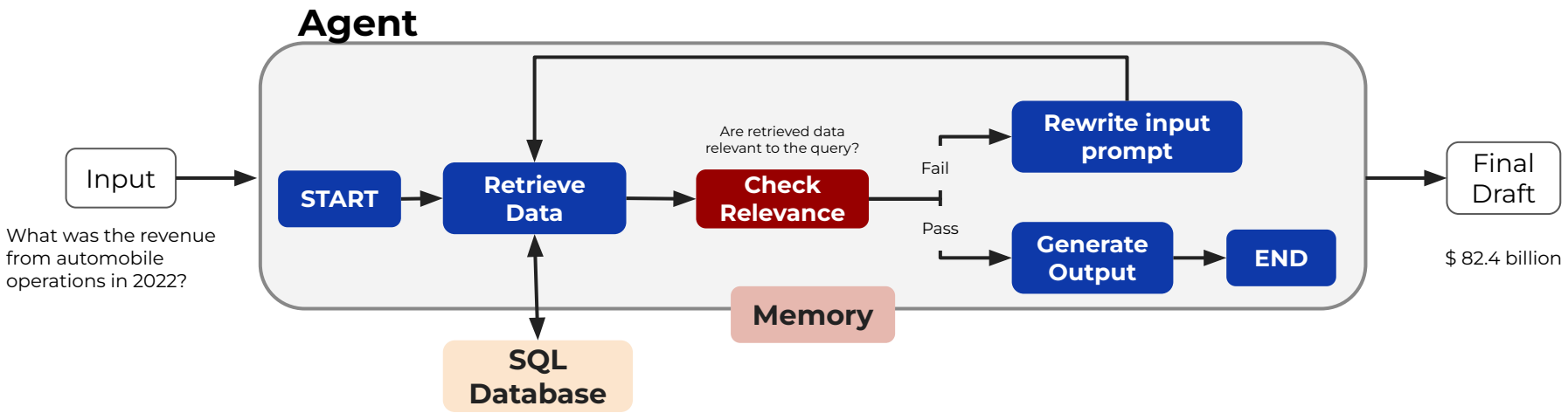
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Self Reflection in AI Agents

Structured, Checkpoint-Driven Workflow: The agent follows an explicit step-by-step pipeline with built-in decision points to evaluate intermediate results and trigger corrective actions when needed.

Integrated Resources & Context: External resources (e.g., vector databases) and internal memory are invoked at predefined stages to support accurate retrieval and maintain context throughout execution.



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